Andrei Dumitrescu

Software Engineer | Java · Spring Boot · TypeScript · Node.js · REST APIs · PostgreSQL

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SUMMARY

Software engineer building backend web services and full-stack products in Java/Spring Boot and TypeScript/Node.js, with React and PostgreSQL. Technical lead on a production AI product: own the Node.js backend and REST APIs, the data and job-queue layer around the LLM service, payments and releases, leading 2 engineers. Previously 3 years of enterprise Java backend, REST/SOAP integration and CI/CD at Ness Digital Engineering, in Scrum/Agile teams (fraud/AML, logistics, tourism).

EXPERIENCE

Technical Lead — ZettaSource July 2025 – September 2026

Product studio — AI companion app (iOS/Android) delivered end to end for a client. Engagement concluded September 2026; available immediately.

- **Led a team of 2 engineers** from architecture to store release — set technical direction, split the design into tasks and held quality through code review.
- **Designed and built the backend** — 80+ TypeScript edge functions (Node.js/Deno) on Supabase/PostgreSQL with REST endpoints, row-level security, Apple/Google/Meta SSO, signed URLs, pg_cron jobs and WebSocket streaming chat — serving 2,800+ registered users.
- **Designed the data model and retry-safe async job queues** around the LLM service (OpenAI, Whisper, ElevenLabs), handling 15,000+ conversations and 95,000+ messages; eliminated duplicate chat replies with a unique index and conflict handler.
- **Implemented subscriptions, paywalls and an in-app coin economy** with RevenueCat and auditable transaction logic, reaching ~3.5% free-to-paid conversion.
- **Built the release pipeline** on GitHub Actions (lint, type checks, sharded tests, migration-sync check, iOS/Android builds, auto-deploy on merge) with 470+ Vitest test files and Maestro E2E flows — ~10 reverts across 450+ PRs.
- **Added operational safety without a dedicated SRE** — feature-flag kill switches, OTA live updates (Capawesome), scheduled health checks on AI-service failure rates and revenue/refunds; PostHog analytics, OneSignal push, A/B assignment and GDPR deletion.
- **Built the React/Vite/Tailwind client with Capacitor** — one codebase for both stores — and owned App Store Connect, TestFlight and Google Play releases.
- **Prototyped Rust payment orchestration** forked from open-source Hyperswitch — PSP connectors, 3DS, authorization/capture, refunds and idempotent retries.

Software Engineer (Freelance, part-time) — ZettaSource June 2024 – July 2025

Evenings & weekends, alongside a full-time role

- **Built and maintained 8 e-commerce storefronts** (Shopify, WooCommerce) for 3 clients; automated inventory and price monitoring, saving ~10 hours/week of manual work.

Software Engineer — Ness Digital Engineering February 2024 – July 2025

- **Implemented business rules and integrations** in Java/Spring Boot on an IBM BAW-type BPM platform for a fraud & anti-money-laundering solution (NICE Actimize) — BPMN 2.0 processes, REST/SOAP APIs, MSSQL/Oracle.
- **Tested releases in staging environments and fixed defects** across BPM processes and integrations until the engagement concluded.
- **Cut regression time from 8–10 hours to ~6 hours** for a US client by refactoring automated suites in Java, Selenium and Cucumber; kept scenarios aligned with new releases and reported defects from failed runs.

Junior Software Engineer — Ness Digital Engineering March 2023 – February 2024

- **Delivered ~5 new backend microservices and contributed to 15+** on a maritime freight-exchange platform (bids, vessel positions) using Spring Boot, Hibernate/JPA and Kafka.
- **Built and maintained 4 CI/CD pipelines** in Jenkins/Groovy — automated version updates and git tagging with jgitver, and investigated build/deploy failures on Docker, Kubernetes, SonarQube and Nexus.

Software Engineer Trainee — Ness Digital Engineering July 2022 – March 2023

- **Built a full-stack tourism web app** for Timișoara with Traian Vuia International Airport via QR codes displayed in the airport — ~200 weekly visitors (Java, MongoDB, JUnit 5, Mockito, Rest Assured).

SKILLS

Languages: Java, TypeScript, JavaScript, SQL, Scala, Rust, Python

Backend: Spring Boot, Node.js, Deno, REST/SOAP APIs, Hibernate/JPA, Kafka, WebSockets, BPM/BPMN 2.0

Frontend & Mobile: React, Vite, Next.js, Tailwind, Capacitor (iOS/Android)

Data: PostgreSQL, Supabase, MSSQL, Oracle, MongoDB

Cloud & DevOps: Docker, Kubernetes, AWS, GitHub Actions, Jenkins, SonarQube, Nexus, Azure DevOps, Linux, Grafana

Testing: JUnit 5, Mockito, Rest Assured, Vitest, Maestro, Selenium, Cucumber

Integrations & AI: OpenAI, Whisper, ElevenLabs, RevenueCat, PostHog, OneSignal, Claude Code, Cursor

EDUCATION

B.Sc. Computer Science — West University of Timișoara, 2020–2023 · Security labs: PortSwigger Web Security Academy, TryHackMe, Hack The Box